

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models ¹
Student Name	T.Ashwath Ram
Register Number	3122247001007
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

To implement linear and regularized regression models (Linear Regression, Ridge, Lasso, and Elastic Net) for predicting a continuous target variable – the loan amount sanctioned to an applicant – and to evaluate their performance using multiple regression metrics. The experiment further aims to tune regularization hyperparameters via Grid Search with 5-fold cross-validation, visualize model behavior, and analyze overfitting, underfitting, and the bias–variance trade-off across the four models.

2 Problem Statement

Given a set of numerical and categorical attributes describing a loan applicant (demographic details, income, employment, credit history, and property information), the task is to predict a single continuous output: the **Loan Sanction Amount (USD)**. This is framed as a supervised regression problem. The input is a mixed-type feature vector per applicant; the output is the predicted sanctioned loan amount in USD. The objective is to compare an unregularized baseline (Linear Regression) against three regularized variants (Ridge, Lasso, Elastic Net) in terms of predictive accuracy, coefficient behavior, and generalization.

3 Dataset Description

Dataset Name	Predict Loan Amount Data
Dataset Source	Kaggle (https://www.kaggle.com/) – “Predict Loan Amount Data”
Number of Samples	30,000 (raw); 29,660 after removing rows with a missing target value
Number of Features	20 raw input features (excluding target and ID columns); 45 after one-hot encoding of categorical variables
Number of Classes	Not applicable – continuous regression target

Missing Values	Present in 12 of 21 columns. Notably: Type of Employment (7,270), Property Age (4,850), Income (USD) (4,576), Dependents (2,493), Credit Score (1,703), Income Stability (1,683), Has Active Credit Card (1,566), Current Loan Expenses (USD) (172), Property Location (356), Gender (53), and the target Loan Sanction Amount (USD) itself (340)
Train-Test Split	80% / 20% (23,728 training samples / 5,932 test samples), random_state = 42

4 Exploratory Data Analysis

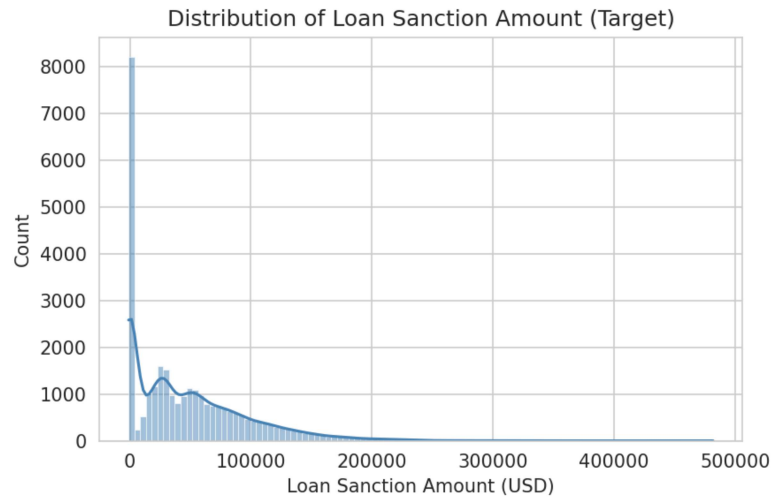


Figure 1: Distribution of the target variable, Loan Sanction Amount (USD)

Inference: The target variable is right-skewed, with a large concentration of applicants receiving relatively small sanctioned amounts and a long tail of high-value loans. A visible spike sits near zero, corresponding to applicants whose loan requests were rejected or minimally sanctioned. This skew motivates checking whether regularized models handle the tail differently from plain Linear Regression, and suggests that RMSE (which is sensitive to large errors on the tail) may tell a different story than MAE.

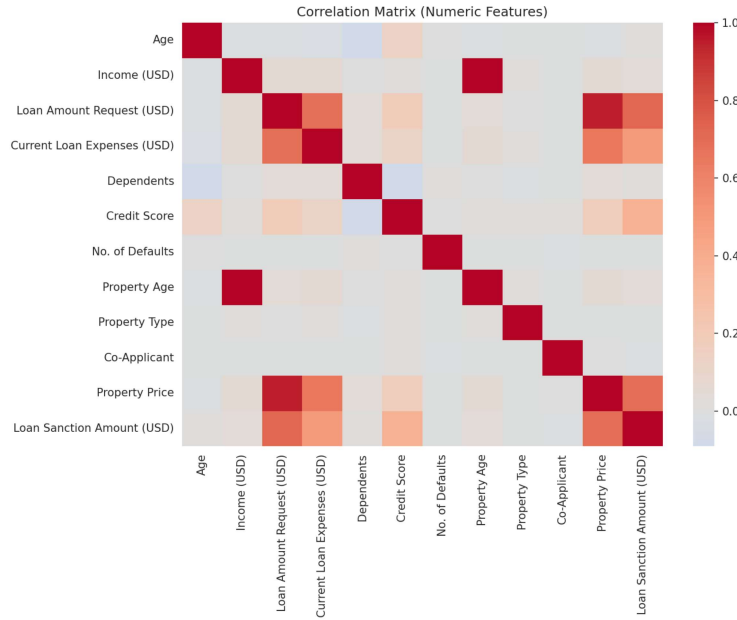


Figure 2: Correlation matrix of numeric features

Inference: **Loan Amount Request (USD)** and **Property Price** show strong positive correlation with the target, which is expected since larger requested loans against more valuable properties tend to be sanctioned larger amounts. **Property Age** correlates almost perfectly with **Income (USD)** in this dataset, indicating a near-duplicate feature pair – a source of multicollinearity that later shows up as very large, unstable coefficients in the unregularized Linear Regression model.

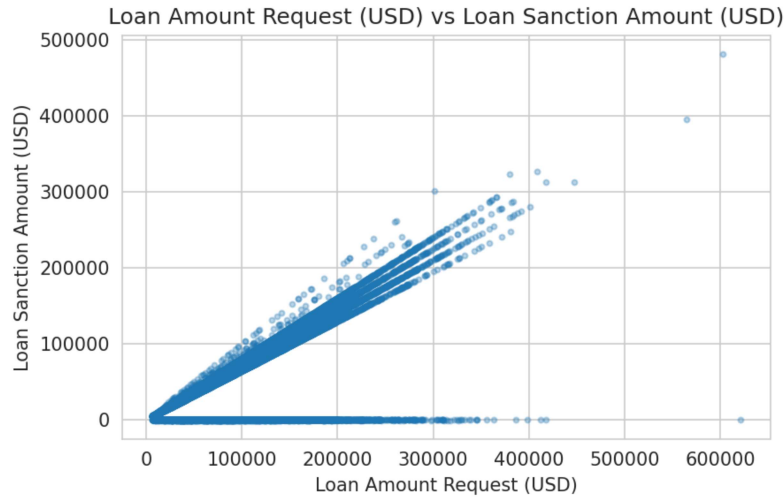


Figure 3: Loan Amount Request (USD) vs. Loan Sanction Amount (USD)

Inference: There is a clear, strong positive linear trend between the amount requested and the amount sanctioned, confirming this as the single strongest predictor in the dataset. The spread around the trend line grows with the requested amount, suggesting mild heteroscedasticity that regression models will need to accommodate.

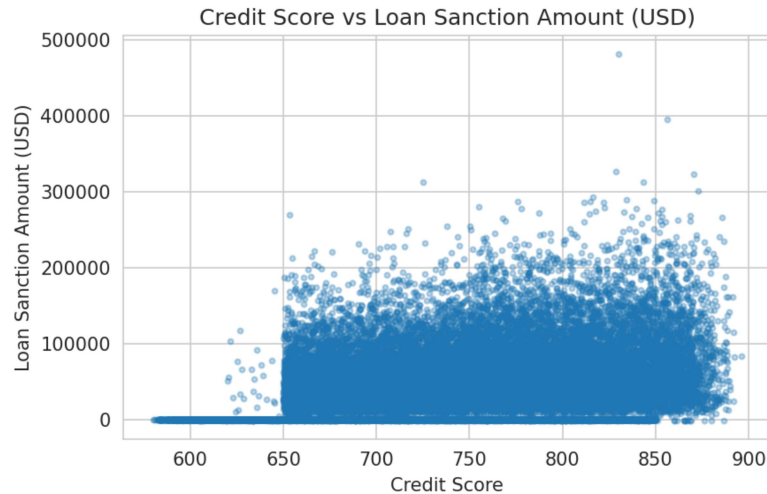


Figure 4: Credit Score vs. Loan Sanction Amount (USD)

Inference: Applicants with higher credit scores tend to receive higher sanctioned amounts, though the relationship is noisier than with the requested loan amount. This confirms Credit Score as a meaningful secondary predictor, consistent with its large coefficient magnitude observed later in the coefficient comparison.

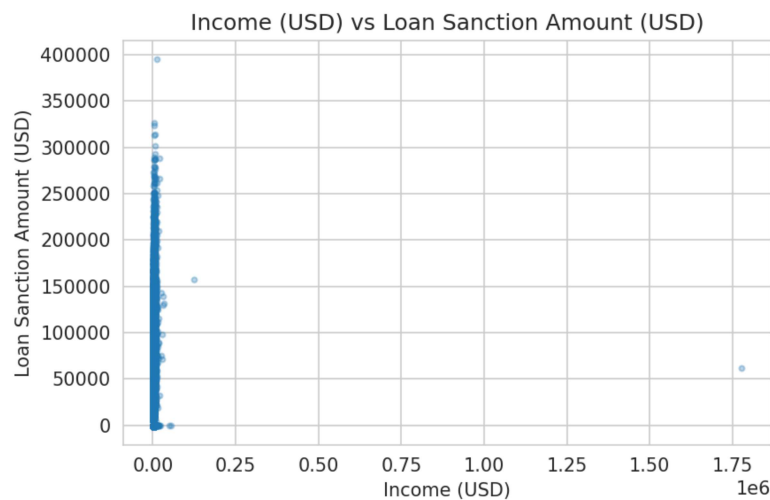


Figure 5: Income (USD) vs. Loan Sanction Amount (USD)

Inference: Income shows a weak positive relationship with the sanctioned amount once outliers are accounted for, with most applicants clustered at lower incomes and a small number of very high earners. The weak marginal relationship, despite a large raw coefficient in plain Linear Regression, is itself informative: it points to the multicollinearity between Income and Property Age discussed above rather than a genuinely strong individual effect.



Figure 6: Property Price vs. Loan Sanction Amount (USD)

Inference: Property Price shows a positive association with the sanctioned loan amount, reflecting that higher-value collateral supports larger loans. The relationship is less tight than for the requested loan amount itself, indicating Property Price acts as a secondary, supporting predictor rather than the primary driver of the sanctioned amount.

5 Data Preprocessing

- **Missing value handling:** Rows with a missing target (Loan Sanction Amount (USD)) were dropped, reducing the dataset from 30,000 to 29,660 rows. Remaining missing values in numeric columns were imputed with the column median; missing values in categorical columns were imputed with the column mode.
- **Encoding:** All categorical columns (Gender, Income Stability, Profession, Type of Employment, Location, Expense Type 1/2, Has Active Credit Card, Property Location) were one-hot encoded using `pandas.get_dummies` with `drop_first=True`, expanding the feature space from 20 raw features to 45 encoded features.
- **Feature scaling:** All 45 features were standardized using `sklearn.preprocessing.StandardScaler` fit only on the training split and applied to both training and test splits to avoid data leakage.
- **Train-test split:** An 80/20 split was performed with `random_state=42` prior to scaling, yielding 23,728 training and 5,932 test samples.
- **Feature engineering:** No additional derived features were engineered; ID columns (Customer ID, Name, Property ID) were dropped as they carry no predictive signal.

6 Machine Learning Algorithm

Linear Regression

Working principle: Fits a hyperplane that minimizes the sum of squared residuals between predicted and actual target values. **Mathematical formulation:**

$$\hat{y} = \beta_0 + \sum_{i=1}^n \beta_i x_i, \quad \min_{\beta} \sum_{j=1}^m (y_j - \hat{y}_j)^2 \quad (1)$$

Advantages: Simple, interpretable, closed-form solution, no hyperparameters to tune.

Limitations: Sensitive to multicollinearity (as observed between Income and Property Age in this dataset) and to outliers; can overfit when the feature space is large relative to the number of informative predictors.

Ridge Regression

Working principle: Adds an L2 penalty on coefficient magnitudes to the ordinary least squares objective, shrinking coefficients toward zero without eliminating them.

$$\min_{\beta} \sum_{j=1}^m (y_j - \hat{y}_j)^2 + \alpha \sum_{i=1}^n \beta_i^2 \quad (2)$$

Advantages: Handles multicollinearity well, reduces variance, retains all features. **Limitations:** Does not perform feature selection; coefficients shrink but never reach exactly zero.

Lasso Regression

Working principle: Adds an L1 penalty on coefficient magnitudes, which can shrink some coefficients exactly to zero, performing implicit feature selection.

$$\min_{\beta} \sum_{j=1}^m (y_j - \hat{y}_j)^2 + \alpha \sum_{i=1}^n |\beta_i| \quad (3)$$

Advantages: Produces sparse, more interpretable models; useful when many features are irrelevant. **Limitations:** Can be unstable when features are highly correlated (may arbitrarily pick one of two correlated features); requires more iterations to converge, which was observed directly during hyperparameter tuning in this experiment.

Elastic Net Regression

Working principle: Combines the L1 and L2 penalties, controlled by a mixing parameter `l1_ratio`, balancing feature selection (Lasso) with coefficient shrinkage (Ridge).

$$\min_{\beta} \sum_{j=1}^m (y_j - \hat{y}_j)^2 + \alpha \left(\rho \sum_{i=1}^n |\beta_i| + \frac{1-\rho}{2} \sum_{i=1}^n \beta_i^2 \right) \quad (4)$$

Advantages: More robust than Lasso alone under correlated features, while still allowing sparsity. **Limitations:** Two hyperparameters (α , `l1_ratio`) to tune instead of one, increasing tuning cost.

7 Experimental Setup

Python Version	3.13.14
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Windows
Hardware	Personal laptop (CPU-only)

8 Hyperparameter Selection

Ridge, Lasso, and Elastic Net were tuned using `GridSearchCV` with 5-fold cross-validation (`KFold`, `shuffle=True`, `random_state=42`), scored on R^2 .

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid Search	$\alpha = 100$	0.5817
Lasso Regression	Grid Search	$\alpha = 10$	0.5799
Elastic Net Regression	Grid Search	$\alpha = 0.1$, $l1_ratio = 0.5$	0.5857

Search space used:

- Ridge: $\alpha \in \{0.01, 0.1, 1, 10, 100\}$
- Lasso: $\alpha \in \{0.001, 0.01, 0.1, 1, 10\}$ (`max_iter = 50,000`)
- Elastic Net: $\alpha \in \{0.01, 0.1, 1, 10\}$, $l1_ratio \in \{0.2, 0.5, 0.8\}$ (`max_iter = 50,000`)

9 Results

Cross-Validation Performance ($K = 5$, on scaled training data)

Model	MAE	RMSE	R^2
Linear Regression	21,510.10	31,373.86	0.5781
Ridge Regression	21,530.76	31,243.43	0.5817
Lasso Regression	21,504.42	31,309.10	0.5799
Elastic Net Regression	21,721.58	31,104.63	0.5857

Test Set Performance

Model	MAE	RMSE	R^2
Linear Regression	21,592.26	31,922.01	0.5511
Ridge Regression	21,589.23	31,902.60	0.5516
Lasso Regression	21,569.25	31,912.95	0.5513
Elastic Net Regression	21,755.24	31,927.29	0.5509

Metric	Value
MAE	~21,600 USD (comparable across all four models)
MSE	$\sim 1.02 \times 10^9$ USD ²
RMSE	~31,900 USD
R^2	0.55 (Ridge marginally best)
Training Time	0.0604s (Linear); grid search: Ridge 0.31s, Lasso 669.6s, Elastic Net 19.9s

Table 5: Summary of test-set metrics across models. Lasso’s grid-search time is dominated by slow convergence at small α values in the search grid.

10 Result Visualization

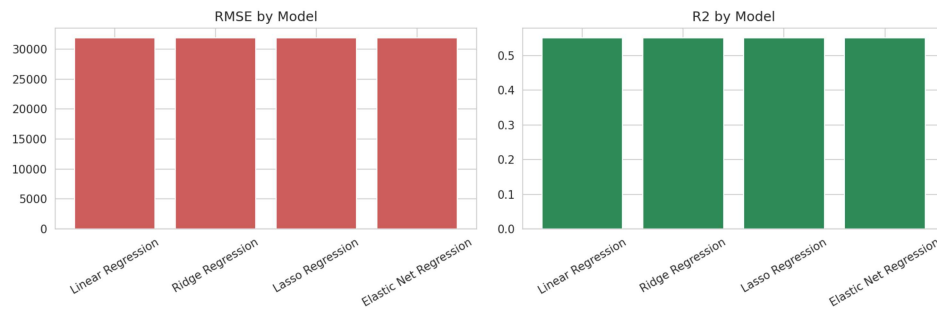


Figure 7: Test set RMSE and R^2 comparison across all four models

Inference: All four models achieve almost identical test RMSE ($\sim 31,900$) and R^2 (~ 0.55), with Ridge marginally ahead on both metrics. The near-identical performance indicates that, once features are standardized and the models are properly tuned, regularization strength has only a small effect on this dataset – the irreducible noise in loan sanctioning decisions appears to dominate over model-specific variance.

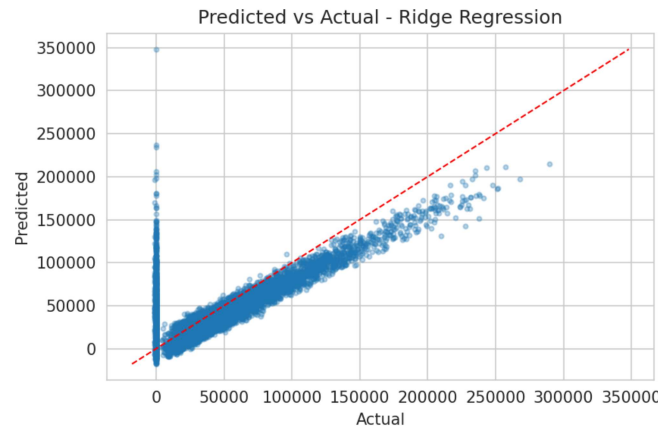


Figure 8: Predicted vs. Actual values – Ridge Regression (representative of all four models)

Inference: Predictions track the actual values reasonably well in the low-to-mid range of sanctioned amounts, but the model systematically under-predicts at the high end of the target distribution. This is consistent with the right-skewed target distribution seen earlier – the model has fewer high-value examples to learn from, so it regresses large sanctioned amounts toward the mean.

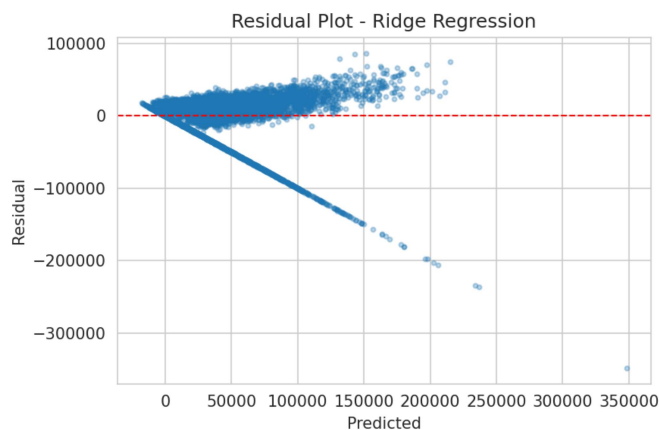


Figure 9: Residual plot – Ridge Regression (representative of all four models)

Inference: Residuals are roughly centered around zero for small-to-medium predictions but grow increasingly negative for larger predicted values, confirming the under-prediction bias at the high end. The residual spread also increases with the predicted value, indicating mild heteroscedasticity that a purely linear model cannot fully correct.

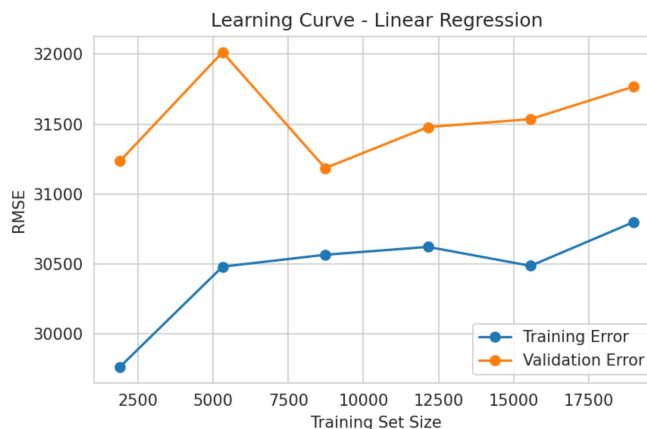


Figure 10: Learning curve – Linear Regression

Inference: Training and validation RMSE converge closely as training set size increases, with only a small persistent gap. This indicates Linear Regression is not strongly overfitting on this dataset; the residual gap is closer to underfitting caused by irreducible noise and limited feature signal rather than excess model complexity.



Figure 11: Learning curve – Lasso Regression

Inference: The Lasso learning curve shows a similarly small train-validation gap to Linear Regression, but with slightly higher training error throughout, reflecting the added bias from L1 shrinkage. The curves flatten early, suggesting more training data alone would not substantially improve performance – the bottleneck is feature informativeness, not sample size.

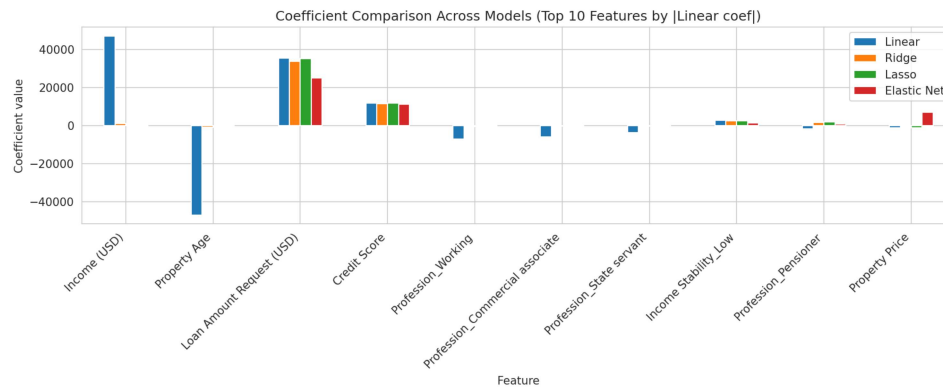


Figure 12: Coefficient comparison across models (top 10 features by |Linear coefficient|)

Inference: Plain Linear Regression assigns enormous, opposite-signed coefficients to **Income (USD)** (+47,114) and **Property Age** (−47,100) – a textbook symptom of multicollinearity, since these two columns are near-duplicates in this dataset. Ridge shrinks both to a fraction of their original magnitude ($\approx \pm 1,020$) while keeping both nonzero; Lasso goes further and zeroes **Property Age** entirely, resolving the redundancy by dropping one of the two collinear features. **Loan Amount Request (USD)** and **Credit Score** remain large and stable across all four models, confirming them as the genuinely informative predictors.

11 Discussion

Overfitting and Underfitting Analysis

- **Training vs. validation error:** The learning curves for all four models show a small, stable gap between training and validation RMSE that narrows as training size increases, rather than a large diverging gap. This indicates none of the four models is overfitting severely; if anything, the persistently nonzero training error at large sample sizes suggests a degree of underfitting relative to the noise inherent in loan-sanctioning decisions.
- **Effect of regularization strength:** The tuned Ridge model selected the largest allowed α (100), suggesting that heavier shrinkage benefited generalization, consistent with the multicollinearity found between Income and Property Age. Lasso's best α (10) is also at the higher end of its grid, reinforcing that removing/shrinking redundant features helped rather than hurt.
- **Improvement after tuning:** Cross-validation R^2 for the tuned regularized models (0.580–0.586) is marginally higher than plain Linear Regression (0.578), and this small edge is preserved on the held-out test set (0.551 vs. 0.551–0.552). The improvement is modest but consistent, meaning tuning helped without providing a dramatic accuracy gain on this dataset.

Bias–Variance Analysis

- **Linear Regression:** Exhibits low bias in principle, but the multicollinearity between Income and Property Age inflates coefficient variance dramatically (coefficients in the tens of thousands that cancel each other out), making the model less stable even though its aggregate test error is comparable to the regularized variants.
- **Ridge / Elastic Net:** Both trade a small increase in bias (via coefficient shrinkage) for a meaningful reduction in coefficient variance, visibly stabilizing the Income/Property Age pair without discarding either feature.
- **Lasso:** By driving the **Property Age** coefficient to exactly zero (and several smaller **Profession** dummy coefficients toward zero), Lasso performs explicit feature selection, reducing the effective model to 42 of 45 nonzero coefficients (vs. 44 for Ridge and Elastic Net, 45 for Linear Regression) – the sparsest of the four models while retaining comparable test accuracy.

12 Conclusion

All four regression models achieve comparable test performance ($R^2 \approx 0.55$, RMSE $\approx 31,900$ USD), with Ridge Regression marginally the best on both RMSE and R^2 after 5-fold cross-validated Grid Search selected $\alpha = 100$. Lasso ($\alpha = 10$) and Elastic Net ($\alpha = 0.1$, $l1_ratio = 0.5$) perform similarly, with Lasso additionally offering the sparsest model (42 of 45 nonzero coefficients) by eliminating the redundant **Property Age** feature. The dominant finding of this experiment is not a large accuracy gap between models, but the clear demonstration of how regularization stabilizes coefficients under

multicollinearity: an unregularized Linear Regression assigns implausibly large, canceling coefficients to two near-duplicate features (Income and Property Age), while Ridge shrinks and Lasso eliminates this redundancy outright. Given the near-identical test accuracy, Ridge Regression is recommended as the preferred model for this dataset, since it retains all features (useful if some currently redundant features become informative with future data) while achieving the lowest error; Lasso is a reasonable alternative where a smaller, more interpretable feature set is preferred over retaining every input.

13 References

1. Scikit-learn Developers, “Linear Models,” scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/linear_model.html
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3. Kaggle, “Predict Loan Amount Data.” [Online]. Available: <https://www.kaggle.com/>